



Instruction Manual for Electrical Technology Laboratory **Department of Electrical Engineering, IIT (ISM),Dhanbad**

Experiment No.:03

VERIFICATION OF MAXIMUM POWER TRANSFER THEOREM



P. K. Nayak
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(P. K. Nayak)
(Prof. In-Charge, E-Tech Lab)

Experiment No.**TITLE: Verification of Maximum Power Transfer Theorem.****OBJECTIVE:** To verify the maximum power transfer theorem experimentally.**APPARATUS USED:**

Serial No.	Name of the Equipment	Rating	Model No. of the Equipment	Makers Name
1.				
2.				
3.				
4.				

THEORY:**A. Maximum Power Transfer Theorem:**

The maximum power transfer theorem states that a load resistance will abstract maximum power from the network when the load resistance is equal to internal resistance. For maximum power transfer, load resistance $R_L = R_i$, where R_i is the internal resistance of the circuit.

The expression of maximum power, $P_{max} = \frac{V^2}{4R_L}$

Where V is the dc voltage applied.

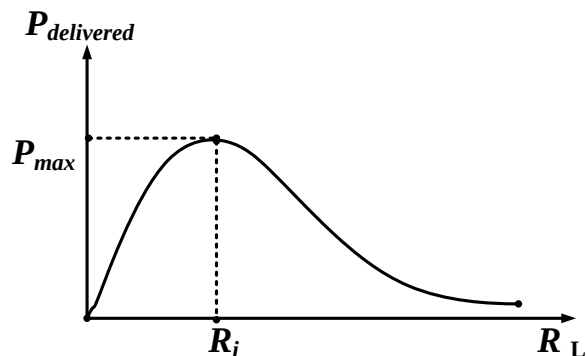
B. Relation Between Power Delivered and Load Resistance:

Fig: 1. Power delivered to load as a function of load resistance R_L .

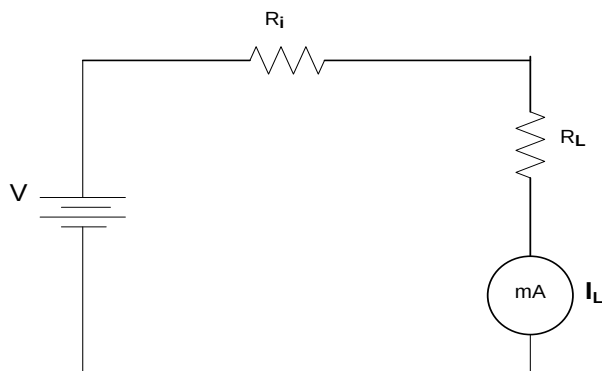
C. Circuit Diagram:

Fig. 2. Circuit diagram of the experimental setup.

PROCEDURE:

1. Connect the circuit diagram as shown in Fig. 1.
2. Take the readings of Voltmeter and Ammeter for different values of R_L .
3. Verify that power is maximum when $R_L = R_i$.

DISCUSSION:

In the network maximum power is transferred when the load resistance is equal to the internal resistance of the network.

PRECAUTIONS:

1. Switch off the supply when not in use.
2. Reading should be taken carefully.
3. All connections should be tight and correctly.

QUESTIONS:

- Q.1. What is load matching?
- Q.2. What is max. power transfer formula?
- Q.3. What is the field of application of this theorem?
- Q.4. What is electric network?
- Q.5. What is necessary to know the polarity of voltage drop across a resistance?
- Q.6. What is the reason that terminal voltage is lesser than emf?
- Q.7. What is the resistance of ideal voltage source?
- Q.8. When will the power extracted from a circuit is maximum?
- Q.9. How the ammeter is connected in circuit?
- Q.10. To find the voltage drop across a resistance, where should the voltmeter be connected?

DATA SHEET

Experiment No. _____

Date: _____

TITLE: Verification of Maximum Power Transfer Theorem

Name: _____ Sem: _____ Year: _____

Adm. No. _____

Experimental Data:-

Maximum Power transfer Theorem:-

Serial No.	Applied voltage (V)	$R_i (\Omega)$	$R_L (\Omega)$	$I_L(\text{mA})$	Power= $I_L^2 \cdot R_L$ (mW)

Signature of the teacher